Machine Learning

Learning Strategies
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Learning Strategies

- **Theoretical** learning strategies (learning theory)
  - When can we learn?
  - How well can we learn?
- **Practical** learning strategies (practical considerations)
  - What is needed?
  - What is possible?

Theoretical Learning Strategies

Learning theory

- In **what cases** can we learn?
- **How well** can we learn?
- **How many examples** do we need to learn?
- How do we **quantify** our ability to **generalize** to unseen data?
- **Which algorithms** are better suited to specific learning **settings**?

Hypothesis Space

- **<u>Hypothesis space *H*</u>**
  - Set of all hypotheses h that the learner may consider
  - ***<u>L</u><u>earning</u> is a <u>search</u> through hypothesis space***

- **Objective**
  - Find h that minimizes some error function w.r.t. training examples
  - Error function example: misclassification

- What about unseen examples?

Generalization

- Recall
  - A **good** hypothesis will **generalize well** (predict unseen examples correctly)
  - Prefer the simplest hypothesis consistent with the data

- Typically
  - If hypothesis h approximates target function ***f*** well over a sufficiently large training set
  - Then h will also approximate **f** well over unobserved examples

- Therefore
  - **A sufficient training set size enables generalization**

- <u>Why do you think that is the case?</u>

Inductive Learning

- **Goal**
  Find an h that agrees with ***f*** on training set
  h is **consistent** if it agrees with ***f*** on all examples

- Finding a consistent hypothesis is not always possible

  - **Insufficient hypothesis space**
    E.g., it is not possible to learn exactly f(x) = ax+b+xsin(x)
    when ***H*** = space of polynomials of finite degree

  - **Noisy data**
    E.g., in weather prediction:
    Identical conditions may lead to rainy and sunny days

Inductive Learning

- A learning problem is **realizable** if the hypothesis space contains the true function — otherwise it is **unrealizable**

- It is difficult to determine whether a learning problem is realizable since the true function is not known

- It is possible to use a very large hypothesis space
  – For example: H = class of all Turing machines

- But there is a **tradeoff** between **expressiveness** of a hypothesis class and the **complexity** of finding a good hypothesis

Practical learning strategies
WHERE DO WE START?

Where do we start?

- We identify a problem
- We want to use Machine Learning to solve it
- Where do we start?
- What questions would you ask?

What is possible? (ML expertise)

- **Model**
  - Problem solved (published)?
  - A model for this problem?
  - Improve an existing model?
  - A different approach?
  - Model complexity?
  - Model implementation/deployment challenges?
  - Is ML necessary/useful?

What is possible? (ML expertise)

- **Data**
  - What data can our model use?
    - Balanced/unbalanced?
    - Size?
    - Available/accessible?
    - Characteristics (modality, multi-source, real-time, dynamic)?
  - How can we preprocess the data?
    - Filter/clean/label/remove noise?
    - Resources available to enable learning at such a data size?
    - Features? Feature Engineering?

What is needed? (Domain expertise)

- Problem?
- Goals? Predict, Classify, Automate?
- Envisioned end result?
- Schedule and budget for deliverables?
- Objectives?
- Constraints?
- Trade-offs?

What is needed?
(Domain expertise)

- **Data**
  - What available domain knowledge can be leveraged?
  - Could data be simulated?
  - What data can we collect?
  - Is it free or do we need to pay for it?

What are the trade-offs? (Decisions & Choices)

- How much accuracy is needed?
- How do we evaluate success?
- What are the computing resources needed/available?
- Will we allow the solution...
  - Unlimited runtime?
  - Unlimited space?

Project Exercise

Identify a potential project topic

- What are some **learning strategies** to consider?